

Module 07: Communication — The Secret Code

Learning Objectives

1. Understand that **Numbers** are symbols we write down.
2. Learn that Math is a **Language** that everyone speaks.
3. Know that symbols let us send messages across time!

Introduction: The Secret Code

Imagine you have 5 apples. You want to tell a friend who lives far away. You can't shout "APPLE!" loud enough. So you write a secret symbol: "**5**". Your friend reads it and knows *exactly* how many you have. Reflecting on the number "5" is **Communication**!

Key Concepts

1. Symbols (The Code)

A symbol is a picture that means something else.

- $\heartsuit$ means Love.
- $+$ means Add.
- 3 means Three things. We agree on the code so we don't get confused.

2. The Universal Language

If you go to France, they say "Cinq." If you go to Spain, they say "Cinco." But everyone writes **5**. Math is a language that the whole world shares!

3. Telling the Future (Writing)

If you write "Meeting at 2:00" on a paper... You are sending a message to your *Future Self*. Writing numbers lets us remember things so we don't forget.

Activities

Activity 1: Alien Numbers

Pretend you are an alien meeting a human. You don't speak English. Show them how many fingers you have (Hold up 3). Draw 3 dots on the ground. Did they understand? Yes! Math works everywhere.

Activity 2: Secret Agent Messages

Write a code:

- A = 1
- B = 2
- C = 3 Send a message: "3 - 1 - 2" (C - A - B -> CAB!). Can your friend decode it?

Summary

Numbers are the most powerful code in the world. They let us share ideas about "How Many" and "How Much" with anyone, anywhere, anytime!

References

- *The History of Counting* by Denise Schmandt-Besserat
- *Swirl by Swirl* by Joyce Sidman (Patterns in nature)